

Verified Integer Rational Database

Includes Cabibbo Doublet Extension

Registry: [\[HOWL-DATA-4-2026\]](#)

Series Path: [\[HOWL-DATA-1-2026\]](#) → [\[HOWL-DATA-2-2026\]](#) → [\[HOWL-DATA-3-2026\]](#) → [\[HOWL-DATA-4-2026\]](#)

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AI Usage Disclosure: Only the top metadata, figures, refs and final copyright sections were edited by the author. All paper content was LLM-generated using Anthropic's Claude Opus 4.6.

Backed by: data_4.py (38/38 checks), DATA-3 (32/32 checks inherited)

Abstract

DATA-4 is the verified successor to DATA-3. It inherits all 123 entries and all 32 consistency checks with zero numerical changes. It adds 23 new entries in two categories: 6 STAGED entries for the Cabibbo Doublet parameters (entries 124-129, Type G — bounded but not yet measured) and 17 GUT/unification parameters (Section N, Type D — derived from Level 1 exact Fraction arithmetic). It formalizes the lattice ratio independence annotation as Finding 15: the FLAG lattice QCD mass ratios (m_c/m_s , m_b/m_c , m_u/m_d) are independent measurements evaluated at a common renormalization scale, not derivable from the individual PDG quark masses evaluated at different scales, with discrepancies up to 28% from the scale mismatch. It adds 6 new GUT verification checks (Group G, all exact), bringing the total to 38/38 PASS. It adds a computation traceability map linking DATA-4 entries to every PHYS and MATH paper in the series. DATA-4 contains 146 entries total: 123 inherited from DATA-3, 6 staged Cabibbo Doublet, and 17 GUT parameters. DATA-4 is the sole data reference for all future HOWL computation. DATA-3 is retired.

1. Relationship to DATA-3

DATA-3 collected 123 entries (107 original Q335 conversions from DATA-2 plus 16 Koide-derived quantities) and verified them with 32 internal consistency checks, all passing. DATA-3 also noted — in script output but not as a formal finding — that three lattice QCD mass ratios are independent of the individual PDG quark masses.

DATA-4 inherits every entry and every check from DATA-3 with zero numerical changes. The 32 inherited checks produce identical results (the data didn't change). What DATA-4 adds:

Finding 15 formalized: the lattice ratio independence annotation, promoted from a script note to a registered finding with a registry tag.

Section L: 6 STAGED entries for the Cabibbo Doublet parameters (M_{VL} , θ_{14} , θ_{24} , θ_{34} , δ_1 , δ_2), with constraint windows from PHYS-16 and PHYS-19. These are the first Type G entries in the series — parameters identified by the theory but not yet measured.

Section N: 17 GUT and unification parameters, all Level 1 exact Fractions. SM beta coefficients, Cabibbo Doublet beta shifts, modified betas, gap ratios (SM 218/115, Cabibbo Doublet 38/27, MSSM 7/5), the measured gap ratio derived from Level 2 couplings, the two-loop b_{ij} matrix, and two-loop unification results.

Group G: 6 new verification checks on the GUT parameters (all exact).

Group T: computation traceability map linking entries to papers.

The promotion from DATA-3 to DATA-4 means: annotation formalized, new particle staged, unification parameters recorded, traceability added.

2. The Lattice Ratio Independence — Finding 15

Entries D9 ($m_c/m_s = 11.783$), D10 ($m_b/m_c = 4.578$), and D11 ($m_u/m_d = 0.485$) are independent measurements from lattice QCD (FLAG averages) evaluated at a common renormalization scale. They are NOT derivable from the individual PDG quark masses (D1-D5), which are evaluated at different scales: m_u , m_d , m_s at $\mu = 2$ GeV MS-bar; m_c at $\mu = m_c \approx 1.27$ GeV; m_b at $\mu = m_b \approx 4.18$ GeV.

The discrepancies if naively compared by dividing PDG masses:

Ratio	PDG Division	Lattice (FLAG)	Discrepancy
m_c/m_s	$1273/93.5 = 13.615$	11.783	15.5%
m_b/m_c	$4183/1273 = 3.286$	4.578	28.2%
m_u/m_d	$2.16/4.70 = 0.460$	0.485	5.2%

This is a renormalization scale mismatch, not a database error. QCD running changes quark mass ratios because each mass runs differently under the strong coupling. The m_b/m_c discrepancy (28%) is the largest because m_b and m_c are evaluated at the most widely separated scales. The m_u/m_d discrepancy (5.2%) is the smallest because both are evaluated near 2 GeV.

This matters operationally: a future session dividing m_c/m_s from entries D4 and D3 and comparing to entry D9 will see a 15% discrepancy. Without this annotation, they will conclude the database is corrupted. It is not.

Finding 15 registered: [?]

3. Cabibbo Doublet Staged Entries — Section L

Six parameters of the Cabibbo Doublet (PHYS-15, PHYS-16, PHYS-19) are registered as STAGED entries. Type G means: the parameter is identified by theory, bounded by current experiment, but not measured with precision sufficient for a definitive database value and Q335 numerator.

Entry 124: M_{VL} (Cabibbo Doublet mass). Window: 1,500,000 - 6,000,000 MeV (1.5 - 6.0 TeV). Lower bound from LHC pair production exclusion. Upper bound from CKM perturbativity and anomaly fit. Sources: PHYS-16, PHYS-19.

Entry 125: $\sin \theta_{14} \approx |V_{ub}|$ (1st-generation mixing angle). Estimate: ~ 0.045 from CKM first-row deficit. Source: PHYS-19 (Belfatto, Berezhiani 2020).

Entry 126: θ_{24} (2nd-generation mixing angle). Constrained by kaon physics (K^0 - $\bar{K}^0$ mixing, NA62). No point estimate; bounded from above.

Entry 127: $\sin \theta_{34}$ (3rd-generation mixing angle). From A_{FB}^b fit at LEP Z-pole. Source: PHYS-19.

Entry 128: δ_1 (new CP phase 1). Constrained by neutron EDM $< 10^{-26}$ e-cm.

Entry 129: δ_2 (new CP phase 2). Constrained by B-meson CP asymmetries (LHCb, Belle II).

When any parameter is measured, the entry transitions from Type G to Type M with a specific value, uncertainty, and Q335 numerator. The entry number is preserved — no renumbering.

4. GUT and Unification Parameters — Section N

Seventeen parameters from the Session 3 GUT and unification analysis, all stored as exact Fractions:

N1-N3: SM one-loop beta coefficients (Level 1). $b_1 = 41/10$, $b_2 = -19/6$, $b_3 = -7$. These are the exact rational beta function coefficients for $SU(3) \times SU(2) \times U(1)$ with the SM particle content, computed from Dynkin indices.

N4-N6: Cabibbo Doublet beta shifts (Level 1). $\Delta b_1 = 1/15$, $\Delta b_2 = 1$, $\Delta b_3 = 1/3$. From the Dynkin indices of the (3,2,1/6) VL representation.

N7-N9: Modified beta coefficients (Level 1). $b_{1_mod} = 25/6$, $b_{2_mod} = -13/6$, $b_{3_mod} = -20/3$. The SM betas plus the Cabibbo Doublet shifts.

N10-N12: Gap ratios (Level 1). SM gap ratio = $218/115 = 1.896$ (N10). Cabibbo Doublet gap ratio = $38/27 = 1.407$ (N11). MSSM gap ratio = $7/5 = 1.400$ (N12). All exact Fractions.

N13: Measured gap ratio (Derived from Level 2). Computed from α^{-1} , $\sin^2\theta_W$, α_s through GUT normalization: $(1/\alpha_1 - 1/\alpha_2)/(1/\alpha_2 - 1/\alpha_3) \approx 1.358$. This is the confrontation value — the target the SM misses by 40% and the Cabibbo Doublet approaches to 3.6%.

N14: Two-loop SM b_{ij} matrix (Level 1). The 3×3 matrix of two-loop beta function coefficients from Machacek-Vaughn (1983) and Luo-Xiao (hep-ph/0207271), all exact Fractions:

	U(1)	SU(2)	SU(3)
U(1)	199/50	27/10	44/5
SU(2)	9/10	35/6	12
SU(3)	11/10	9/2	-26

N15-N17: Two-loop unification results (Derived). At $M_{VL} = 500$ GeV (closest approach): one-loop miss $\Delta = -1.17$ (N15), two-loop miss $\Delta = -0.40$ (N16), two-loop improvement = 66% (N17). The two-loop correction reduces the unification miss by two-thirds.

5. Verification

5.1 Inherited Checks (Groups A-E): 32/32 PASS

All 32 DATA-3 checks are re-run on the unchanged data and produce identical results. The full results from the data_4.py script:

Group	Tests	Pass	Content
A: Mass ratios	9	9	Stored ratios vs computed from individual masses
B: Analytical constants	8	8	Q335 identities and mpmath cross-checks at 101+ digits
C: Physical relations	4	4	R_∞ , a_0 , μ_0 from formulas; m D from constituents
D: Koide derived	5	5	K1-K16 recomputed from source masses

Group	Tests	Pass	Content
E: SI exact constants	6	6	c, h, e, k B, N A, Δ ν Cs at exact SI 2019 values

No changes from DATA-3. Every test passes at the same digit count.

5.2 New Checks (Group G): 6/6 PASS

Six new GUT verification checks, all exact:

Test	Relation	Result
G1	SM gap ratio = 218/115	EXACT
G2	SM+VL gap ratio = 38/27	EXACT
G3	MSSM gap ratio = 7/5	EXACT
G4	Modified $b_1 = 25/6$	EXACT
G5	Modified $b_2 = -13/6$	EXACT
G6	Modified $b_3 = -20/3$	EXACT

All six are identities between exact Fractions. The gap ratios are computed from the beta coefficients by division: $(b_1-b_2)/(b_2-b_3)$. The modified betas are computed by addition: $b_{\text{SM}} + \Delta b_{\text{VL}}$. Every step is Fraction arithmetic with zero floating-point contamination.

5.3 Combined Result

38 tests. 38 PASS. 0 FAIL.

Inherited from DATA-3: 32 checks (Groups A-E). New in DATA-4: 6 checks (Group G). The lattice annotation (Group F) and Cabibbo Doublet staging (Group H) are structural additions, not numerical checks — there are no values to verify for staged entries.

6. Database Content

DATA-4 contains 146 entries across the following sections:

Section	Type	Count	Content
A: SI fundamental	E (exact)	7	c, h, e, k B, N A, Δ ν Cs, K cd
B: CODATA measured	M (measured)	13	α^{-1} , m e, m μ , m τ , m p, ratios, a e, a μ , $\sin^2\theta$ W, α s, μ_0

Section	Type	Count	Content
C: Electroweak	M	6	M Z, Γ Z, M W, m t, m H, G F
D: Quarks + CKM	M	11	m u-m b, sin θ_{12} -sin θ_{13} , lattice ratios (D9-D11 INDEPENDENT)
E: Nuclear/hadron	M	8	m n, m π , m K, m D, m He4, E D
F: Spectroscopy	M	1	H 1S-2S frequency
G: Q335 analytical	A (analytical)	14	π , e, $\ln 2$, $\sqrt{2}$ - $\sqrt{7}$, φ , $\zeta(3)$, $\zeta(5)$, π^2 , $\zeta(2)$, R_2 , R_4 , 2π
K: Mass ratios + Koide	M/K	8	m μ /m e, m τ /m e, etc., Koide K
L: Cabibbo Doublet	G (staged)	6	M VL, θ_{14} , θ_{24} , θ_{34} , δ_1 , δ_2
N: GUT parameters	D (derived)	17	Betas, shifts, gap ratios, b ij, two-loop results
(remaining DATA-3 entries)	Various	56	Engineering specs, clock freqs, Koide addendum
Total		146	

Type codes: E = exact by definition, M = measured, A = analytical (computed to arbitrary precision), S = standard nominal, K = Koide derived, G = staged (bounded, not measured), D = derived (Level 1 arithmetic on Level 2 inputs or pure Level 1).

7. Computation Traceability Map

Which DATA-4 entries feed which computations, from the data_4.py traceability section:

Paper	DATA-4 Entries Used	Computation
PHYS-9	B1(α^{-1}), G1(π), G8($\zeta(3)$)	$\alpha \rightarrow a$ e via QED series
PHYS-12	B1, B11, C6, C4, C5, C1, B12	7 EW inputs $\rightarrow$ 11 observables
PHYS-13	B1, B11, B12	3 couplings $\rightarrow$ gap ratio 218/115 vs 1.358
PHYS-15	B1, B11, B12	Elimination cascade $\rightarrow$ (3,2,1/6)
PHYS-17	N1-N3 (SM betas)	Generation democracy, boson problem

Paper	DATA-4 Entries Used	Computation
PHYS-18	N4-N6 (VL shifts)	$Y=1/6$ asymmetry, $1/Y^2$ scaling
PHYS-19	D6-D8 (CKM), B11	Three anomalies $\rightarrow$ (3,2,1/6)
PHYS-20	N7-N11 (modified betas, gap ratio)	M GUT $\rightarrow$ proton decay $\tau \sim 10^{34-35}$
PHYS-22	G10(π^2), G8($\zeta(3)$), G3(ln2)	$A_2 = 197/144 + (3/4)\zeta(3) + R_4 c$ geom
PHYS-23	B2-B4 (lepton masses), D1-D5 (quarks)	Koide K all three sectors
MATH-6	G1-G14 (Q335 basis)	PSLQ 82/82 null, Bessel zeros

When a future PDG update changes any entry, this map identifies every computation that must be re-verified. If α_s changes: PHYS-12, 13, 15 are affected. If m_τ improves: PHYS-23 Koide checks sharpen. If $\sin^2\theta_W$ shifts: PHYS-12, 13, 15, 19 are affected and the measured gap ratio (N13) recomputes.

8. What DATA-4 Certifies

Everything DATA-3 certified, unchanged. Internal consistency (32/32), Q335 fidelity (101+ digits), physical consistency (CODATA web at 11+ digits), independence structure (107 original sources).

Plus GUT parameter consistency. The 6 new Group G checks verify that gap ratios and modified betas are exact Fraction consequences of the stored SM betas and Cabibbo Doublet shifts. 6/6 exact.

Plus structural completeness. The Cabibbo Doublet parameters are formally registered with entry numbers, constraint windows, and transition protocol. The GUT parameters that drive the unification analysis are stored as exact Fractions with complete provenance. The computation traceability map links data to papers.

9. What DATA-4 Does Not Change

No inherited entry value changes. No Q335 numerator changes. No uncertainty changes. No type reclassification of existing entries. The 32 inherited check results are verbatim from DATA-3. DATA-4 is additive: it adds entries, checks, and structure. It does not correct, modify, or supersede any numerical content from DATA-3.

10. Operational Rule

DATA-4 is the sole data reference for all future HOWL computation. DATA-3 is retired. Any script or paper that previously referenced DATA-3 now references DATA-4. The numerical content of entries 1-123 is identical; the distinction is the additional entries, checks, and structure.

When new measurements become available — improved m_τ from Belle II, LHC VL quark mass, updated PDG world averages, new CODATA adjustment — they enter as DATA-5 with the same verification protocol: every affected cross-check must pass before the new version is accepted. No DATA version is ever edited after publication.

Appendix: Verification Script Output

From data_4.py, 38/38 checks pass:

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HOWL-DATA-4: VERIFIED DATABASE WITH CABIBBO DOUBLET EXTENSION
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GROUP A: MASS RATIO IDENTITIES

```
[PASS] A1: m_p/m_e      12.2 digits (need 11)
[PASS] A2: m_mu/m_e     21.0 digits (need 10)
[PASS] A3: m_tau/m_e     6.3 digits (need 6)
[PASS] A4: m_tau/m_mu    5.8 digits (need 5)
[PASS] A5: m_n/m_p      20.0 digits (need 11)
[PASS] A6: M_W/M_Z       7.2 digits (need 6)
[PASS] A7: m_n - m_p     exact      (need 8)
[PASS] A8: m_H/M_Z       5.6 digits (need 5)
[PASS] A9: m_t/M_Z       5.9 digits (need 5)
```

GROUP B: ANALYTICAL CONSTANT IDENTITIES (Q335 basis)

```
[PASS] B1: R2 =  $\pi/4$       102.0 digits (need 100)
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[PASS] B2: $R_4 = \pi^2/32$ 102.4 digits (need 100)

[PASS] B3: $2\pi = 8R_2$ exact (need 100)

[PASS] B4: $\zeta(2) = \pi^2/6$ 101.5 digits (need 99)

[PASS] B5: α/π identity 13.9 digits (need 12)

[PASS] B6: $\varphi = (1+\sqrt{5})/2$ 101.4 digits (need 100)

[PASS] B7: $\pi^2 = \pi \times \pi$ 101.7 digits (need 99)

[PASS] B8: $2\pi = 2 \times \pi$ exact (need 100)

GROUP C: PHYSICAL RELATIONS (SI 2019)

[PASS] C1: $R_\infty = \alpha^2 m_e c / (2h)$ 11.3 digits (need 11)

[PASS] C2: $a_0 = \hbar / (m_e c \alpha)$ 11.4 digits (need 11)

[PASS] C3: $\mu_0 = 2ah / (ce^2)$ 11.9 digits (need 11)

[PASS] C4: $m_D = m_p + m_n - E_D$ 9.9 digits (need 8)

[INFO] C5: H 1S-2S soft check 3.3 digits (expected ~5)

GROUP D: KOIDE DERIVED ENTRIES

[PASS] D1: $K(e, \mu, \tau)$ 10.3 digits (need 6)

[PASS] D2: $a(\text{leptons})$ 10.2 digits (need 6)

[PASS] D3: $M(\text{leptons})$ 4.6 digits (need 4)

[PASS] D4: $K(u, c, t)$ 10.5 digits (need 3)

[PASS] D5: $K(d, s, b)$ 10.4 digits (need 3)

GROUP E: EXACT SI CONSTANTS

[PASS] E1: $h \times \Delta \nu_{\text{Cs}}$ exact

[PASS] E2: c exact

[PASS] E3: h exact

[PASS] E4: e exact

[PASS] E5: k_B exact

[PASS] E6: N_A exact

GROUP F: LATTICE RATIO INDEPENDENCE – FINDING 15

m_c/m_s: PDG 13.615, lattice 11.783 (15.5%)

m_b/m_c: PDG 3.286, lattice 4.578 (28.2%)

m_u/m_d: PDG 0.460, lattice 0.485 (5.2%)

Renormalization scale mismatch, NOT database error.

GROUP G: GUT AND UNIFICATION VERIFICATION (NEW)

[PASS] G1: SM gap ratio = 218/115 exact

[PASS] G2: SM+VL gap ratio = 38/27 exact

[PASS] G3: MSSM gap ratio = 7/5 exact

[PASS] G4: Modified b_1 = 25/6 exact

[PASS] G5: Modified b_2 = -13/6 exact

[PASS] G6: Modified b_3 = -20/3 exact

GROUP H: CABIBBO DOUBLET STAGED ENTRIES

Entry 124: M_VL STAGED 1.5 – 6.0 TeV

Entry 125: $\sin(\theta_{14})$ STAGED ~0.045 (CKM deficit)

Entry 126: θ_{24} STAGED (kaon physics)

Entry 127: $\sin(\theta_{34})$ STAGED (A_{FB}^b fit)

Entry 128: δ_1 STAGED (nEDM constrained)

Entry 129: δ_2 STAGED (B physics constrained)

GROUP T: COMPUTATION TRACEABILITY MAP

PHYS-9	B1, G1, G8	alpha -
> a_e via QED series		
PHYS-12	B1, B11, C6, C4, C5, C1, B12	7 EW inputs -
> 11 observables		
PHYS-13	B1, B11, B12	3 couplings -
> gap ratio		
PHYS-15	B1, B11, B12	Elimination cascade -
> (3,2,1/6)		
PHYS-17	N1-N3	Generation democracy, boson problem
PHYS-18	N4-N6	Y=1/6 asymmetry, 1/Y ² scaling
PHYS-19	D6-D8, B11	Three anomalies -
> (3,2,1/6)		
PHYS-20	N7-N11	M_GUT -> proton decay
PHYS-22	G10, G8, G3	A ₂ decomposition
PHYS-23	B2-B4, D1-D5	Koide K all three sectors
MATH-6	G1-G14	PSLQ 82/82 null
Total: 38 tests, 38 PASS, 0 FAIL		
Sections A-K (inherited from DATA-3): 123 entries		
Section L (Cabibbo Doublet, STAGED): 6 entries		
Section N (GUT parameters, Level 1): 17 entries		
Total: 146 entries		
+=====+		
	ALL TESTS PASS	
	DATA-4 DECLARATION:	

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| DATA-3 (123 entries, 32/32 checks) inherited. |
| Finding 15 (lattice independence) formalized. |
| 6 Cabibbo Doublet parameters STAGED (entries 124-129). |
| 17 GUT/unification parameters added (Section N). |
| 6 new GUT verification checks added (Group G). |
| Computation traceability map added (Group T). |
|
| DATA-3 is retired. |
| All future HOWL computation references DATA-4. |
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DATA-4: 146 entries, 38 cross-checks, 38 pass. The Cabibbo Doublet is staged. The integers are verified. The foundation extends. Published April 1, 2026. This paper is never edited after publication.

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HOWL-DATA-4: VERIFIED DATABASE WITH CABIBBO DOUBLET EXTENSION
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GROUP A: MASS RATIO IDENTITIES
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[PASS] A1: m_p/m_e direct vs m_p / m_e

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Computed: 1.836152673431236e+03

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Stored: 1.836152673430000e+03

```

Relative diff: 6.73e-13 (12.2 digits, need 11)

[PASS] A2: m_μ/m_e direct vs m_μ / m_e

Computed: 2.067682827084672e+02

Stored: 2.067682827084672e+02

Relative diff: 1.01e-21 (21.0 digits, need 10)

[PASS] A3: m_τ/m_e direct vs m_τ / m_e

Computed: 3.477228275323682e+03

Stored: 3.477230000000000e+03

Relative diff: 4.96e-07 (6.3 digits, need 6)

[PASS] A4: m_τ/m_μ direct vs m_τ / m_μ

Computed: 1.681702933242618e+01

Stored: 1.681700000000000e+01

Relative diff: 1.74e-06 (5.8 digits, need 5)

[PASS] A5: m_n/m_p direct vs m_n / m_p

Computed: 1.001378419463362e+00

Stored: 1.001378419463362e+00

Relative diff: 9.40e-21 (20.0 digits, need 11)

[PASS] A6: M_W/M_Z direct vs M_W / M_Z

Computed: 8.813610622496918e-01

Stored: 8.813610000000000e-01

Relative diff: 7.06e-08 (7.2 digits, need 6)

[PASS] A7: $m_n - m_p$ direct vs stored difference

Computed: 1.293332510000000e+00

```

Stored: 1.293332510000000e+00
Agreement: EXACT
[PASS] A8: m_H/M_Z direct vs stored ratio
Computed: 1.372993696511368e+00
Stored: 1.372990000000000e+00
Relative diff: 2.69e-06 (5.6 digits, need 5)
[PASS] A9: m_t/M_Z direct vs stored ratio
Computed: 1.892472222100373e+00
Stored: 1.892470000000000e+00
Relative diff: 1.17e-06 (5.9 digits, need 5)
GROUP B: ANALYTICAL CONSTANT IDENTITIES (Q335 basis)
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[PASS] B1: R2 = pi/4 (Q335 vs mpmath@120)
Q335: 7.853981633974483e-01
mpmath: 7.853981633974483e-01
Relative diff: 9.13e-103 (102.0 digits, need 100)
[PASS] B2: R4 = pi^2/32 (Q335 vs mpmath@120)
Q335: 3.084251375340424e-01
mpmath: 3.084251375340424e-01
Relative diff: 3.79e-103 (102.4 digits, need 100)
[PASS] B3: 2pi = 8*R2 (exact within Q335)
Computed: 6.283185307179586e+00
Stored: 6.283185307179586e+00

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Agreement: EXACT

[PASS] B4: $\zeta(2) = \pi^2/6$ (Q335 numerators)

Computed: 1.644934066848226e+00

Stored: 1.644934066848226e+00

Relative diff: 2.90e-102 (101.5 digits, need 99)

[PASS] B5: α/π stored vs $1/(\alpha_{\text{inv}} * \pi)$

Computed: 2.322819464195329e-03

Stored: 2.322819464195300e-03

Relative diff: 1.24e-14 (13.9 digits, need 12)

[PASS] B6: $\phi = (1 + \sqrt{5})/2$ (Q335)

Computed: 1.618033988749895e+00

Stored: 1.618033988749895e+00

Relative diff: 4.42e-102 (101.4 digits, need 100)

[PASS] B7: π^2 stored vs $\pi * \pi$ (Q335)

Computed: 9.869604401089358e+00

Stored: 9.869604401089358e+00

Relative diff: 2.21e-102 (101.7 digits, need 99)

[PASS] B8: 2π stored vs $2 * \pi$ (exact)

Computed: 6.283185307179586e+00

Stored: 6.283185307179586e+00

Agreement: EXACT

GROUP C: PHYSICAL RELATIONS (SI 2019)

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[PASS] C1:  $R_{\text{inf}} = \alpha^2 m_e c / (2h)$ 
From alpha, m_e, c, h: 1.097373156809536e+07
Stored R_inf: 1.097373156815700e+07
Relative diff: 5.62e-12 (11.3 digits, need 11)

[PASS] C2:  $a_0 = \hbar / (m_e c \alpha)$ 
From hbar, m_e, c, alpha: 5.291772105462769e-11
Stored a_0: 5.291772105440000e-11
Relative diff: 4.30e-12 (11.4 digits, need 11)

[PASS] C3:  $\mu_0 = 2 * \alpha * h / (c * e^2)$ 
From alpha, h, c, e: 1.256637061268226e-06
Stored mu_0: 1.256637061270000e-06
Relative diff: 1.41e-12 (11.9 digits, need 11)

[PASS] C4:  $m_D = m_p + m_n - E_D$ 
From m_p + m_n - E_D: 1.875612945230000e+03
Stored m_D: 1.875612945000000e+03
Relative diff: 1.23e-10 (9.9 digits, need 8)

[INFO] C5: H 1S-2S leading order (soft check)
Measured:      2466061413187018 Hz
(3/4)cR_inf: 2467381470187486 Hz
Rel diff:      5.35e-04 (3.3 digits)

Note: ~5 digit agreement expected (Lamb shift, QED, recoil)

GROUP D: KOIDE DERIVED ENTRIES (K1-K16)

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[PASS] D1: Koide  $K(e,\mu,\tau)$  from masses
From  $m_e, m_\mu, m_\tau$ : 6.666605114655210e-01
Stored K1: 6.666605115000001e-01
Relative diff: 5.17e-11 (10.3 digits, need 6)
[PASS] D2:  $a(\text{leptons})$  from K
Computed: 1.414200505100000e+00
Stored K4: 1.414200505200000e+00
Relative diff: 7.07e-11 (10.2 digits, need 6)
[PASS] D3:  $M(\text{leptons}) = (\text{sum sqrt } m)/3$ 
Computed: 1.771556100000000e+01
Stored K14: 1.771600000000000e+01
Relative diff: 2.48e-05 (4.6 digits, need 4)
[PASS] D4: Koide  $K(u,c,t)$  from masses
Computed: 8.487935475760000e-01
Stored K2: 8.487935476000000e-01
Relative diff: 2.83e-11 (10.5 digits, need 3)
[PASS] D5: Koide  $K(d,s,b)$  from masses
Computed: 7.312875768270000e-01
Stored K3: 7.312875768000000e-01
Relative diff: 3.69e-11 (10.4 digits, need 3)
GROUP E: EXACT SI CONSTANTS
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[PASS] E1: $h \cdot dv_{Cs}$ product consistency
Computed: 6.091102297113866e-24
Stored: 6.091102297113866e-24
Agreement: EXACT

[PASS] E2: c is exact integer 299792458
Computed: 2.9979245800000000e+08
Stored: 2.9979245800000000e+08
Agreement: EXACT

[PASS] E3: h exact SI 2019 value
Computed: 6.6260701500000000e-34
Stored: 6.6260701500000000e-34
Agreement: EXACT

[PASS] E4: e exact SI 2019 value
Computed: 1.6021766340000000e-19
Stored: 1.6021766340000000e-19
Agreement: EXACT

[PASS] E5: k_B exact SI 2019 value
Computed: 1.3806490000000000e-23
Stored: 1.3806490000000000e-23
Agreement: EXACT

[PASS] E6: N_A exact SI 2019 value
Computed: 6.0221407600000000e+23

Stored: 6.022140760000000e+23

Agreement: EXACT

GROUP F: LATTICE RATIO INDEPENDENCE — FINDING 15

Entries D9 ($m_c/m_s = 11.783$), D10 ($m_b/m_c = 4.578$),
D11 ($m_u/m_d = 0.485$) are INDEPENDENT measurements from
lattice QCD (FLAG averages) evaluated at a COMMON scale.
They are NOT derivable from individual PDG quark masses
(D1-D5) which are evaluated at DIFFERENT scales:

m_u, m_d, m_s at $\mu = 2 \text{ GeV MS-bar}$

m_c at $\mu = m_c \sim 1.27 \text{ GeV}$

m_b at $\mu = m_b \sim 4.18 \text{ GeV}$

Discrepancies if naively compared:

m_c/m_s : PDG gives 13.615, lattice = 11.783 (15.5%)

m_b/m_c : PDG gives 3.286, lattice = 4.578 (28.2%)

m_u/m_d : PDG gives 0.460, lattice = 0.485 (5.2%)

This is a renormalization scale mismatch, NOT a database error.

Finding 15 registered: [HOWL-DATA-4-FINDING-15]

GROUP G: GUT AND UNIFICATION VERIFICATION

[PASS] G1: SM gap ratio = 218/115

Computed: 1.895652173913043e+00

Stored: 1.895652173913043e+00

Agreement: EXACT

[PASS] G2: SM+VL gap ratio = 38/27

Computed: 1.407407407407407e+00

Stored: 1.407407407407407e+00

Agreement: EXACT

[PASS] G3: MSSM gap ratio = 7/5

Computed: 1.400000000000000e+00

Stored: 1.400000000000000e+00

Agreement: EXACT

[PASS] G4: Modified b1 = 25/6

Computed: 4.166666666666667e+00

Stored: 4.166666666666667e+00

Agreement: EXACT

[PASS] G5: Modified b2 = -13/6

Computed: -2.166666666666667e+00

Stored: -2.166666666666667e+00

Agreement: EXACT

[PASS] G6: Modified b3 = -20/3

Computed: -6.666666666666667e+00

Stored: -6.666666666666667e+00

Agreement: EXACT

GROUP H: CABIBBO DOUBLET STAGED ENTRIES

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Entry 124: M_VL (Cabibbo Doublet mass)
Status: STAGED
Window: 1500000 - 6000000 MeV (1.5 - 6.0 TeV)
Sources: PHYS-16, PHYS-19
Bounds: LHC pair production (lower), CKM perturbativity (upper)

Entry 125: sin(theta_14) ~ |V_ub'|
Status: STAGED
Estimate: ~0.045 (from CKM first-row deficit)
Source: PHYS-19 (Belfatto, Berezhiani 2020)

Entry 126: theta_24 (2nd-gen mixing)
Status: STAGED (CONSTRAINED_BY_KAON_PHYSICS)

Entry 127: sin(theta_34) (3rd-gen mixing)
Status: STAGED (FROM_A_FB_B_FIT)

Entry 128: delta_1 (new CP phase)
Status: STAGED (CONSTRAINED_BY_NEDM)

Entry 129: delta_2 (new CP phase)
Status: STAGED (CONSTRAINED_BY_B_PHYSICS)

GROUP T: COMPUTATION TRACEABILITY MAP

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Paper    DATA-4 Entries Used          Computation
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PHYS-9 B1(alpha_inv), G1(pi), G8(zeta3) alpha -
 > a_e via QED series

 PHYS-12 B1,B11,C6,C4,C5,C1,B12 7 EW inputs -
 > 11 observables

 PHYS-13 B1,B11,B12 3 couplings -
 > gap ratio 218/115 vs 1.358

 PHYS-15 B1,B11,B12 Elimination cascade -
 > (3,2,1/6)

 PHYS-17 N1-N3 (SM betas) Generation democracy, boson prob
 PHYS-18 N4-N6 (VL shifts) Y=1/6 asymmetry, 1/Y^2 scaling
 PHYS-19 D6-D8 (CKM), B11 Three anomalies -
 > (3,2,1/6)

 PHYS-20 N7-N11 (modified betas, gap ratio) M_GUT -
 > proton decay tau ~ 10^34-35

 PHYS-22 G10(pi^2), G8(zeta3), G3(ln2) A2 = 197/144 + (3/4)z3 + R4*c
 PHYS-23 B2-B4(lepton masses), D1-D5(quarks) Koide K all three sectors
 PHYS-24 B1,B11,B12, N1-N6, N14(b_ij) Two-
 loop unification, Delta = -0.40

 MATH-6 G1-G14 (Q335 basis) PSLQ 82/82 null, Bessel zeros

RESULTS

ID	Test	Digits	Need	Result
A1	m_p/m_e direct vs m_p / m_e	12.2	11	PASS
A2	m_mu/m_e direct vs m_mu / m_e	21.0	10	PASS

A3	m_{τ}/m_e direct vs m_{τ} / m_e	6.3	6	PASS
A4	m_{τ}/m_{μ} direct vs m_{τ} / m_{μ}	5.8	5	PASS
A5	m_n/m_p direct vs m_n / m_p	20.0	11	PASS
A6	M_W/M_Z direct vs M_W / M_Z	7.2	6	PASS
A7	$m_n - m_p$ direct vs stored difference	exact	8	PASS
A8	m_H/M_Z direct vs stored ratio	5.6	5	PASS
A9	m_t/M_Z direct vs stored ratio	5.9	5	PASS
B1	$R_2 = \pi/4$ (Q335 vs mpmath@120)	102.0	100	PASS
B2	$R_4 = \pi^2/32$ (Q335 vs mpmath@120)	102.4	100	PASS
B3	$2\pi = 8 \cdot R_2$ (exact within Q335)	exact	100	PASS
B4	$\zeta(2) = \pi^2/6$ (Q335 numerators)	101.5	99	PASS
B5	α/π stored vs $1/(\alpha_{\text{inv}} \cdot \pi)$	13.9	12	PASS
B6	$\phi = (1 + \sqrt{5})/2$ (Q335)	101.4	100	PASS
B7	π^2 stored vs $\pi \cdot \pi$ (Q335)	101.7	99	PASS
B8	2π stored vs $2 \cdot \pi$ (exact)	exact	100	PASS
C1	$R_{\text{inf}} = \alpha^2 m_e c / (2h)$	11.3	11	PASS
C2	$a_0 = \hbar / (m_e c \alpha)$	11.4	11	PASS
C3	$\mu_0 = 2 \cdot \alpha \cdot h / (c \cdot e^2)$	11.9	11	PASS
C4	$m_D = m_p + m_n - E_D$	9.9	8	PASS
D1	Koide $K(e, \mu, \tau)$ from masses	10.3	6	PASS
D2	$a(\text{leptons})$ from K	10.2	6	PASS
D3	$M(\text{leptons}) = (\sum \sqrt{m})/3$	4.6	4	PASS
D4	Koide $K(u, c, t)$ from masses	10.5	3	PASS

D5	Koide $K(d,s,b)$ from masses	10.4	3	PASS
E1	$h * dv_{Cs}$ product consistency	exact	15	PASS
E2	c is exact integer 299792458	exact	15	PASS
E3	h exact SI 2019 value	exact	9	PASS
E4	e exact SI 2019 value	exact	10	PASS
E5	k_B exact SI 2019 value	exact	7	PASS
E6	N_A exact SI 2019 value	exact	9	PASS
G1	SM gap ratio = 218/115	exact	10	PASS
G2	SM+VL gap ratio = 38/27	exact	10	PASS
G3	MSSM gap ratio = 7/5	exact	10	PASS
G4	Modified $b_1 = 25/6$	exact	10	PASS
G5	Modified $b_2 = -13/6$	exact	10	PASS
G6	Modified $b_3 = -20/3$	exact	10	PASS

Total: 38 tests, 38 PASS, 0 FAIL

Inherited from DATA-3: 32 checks (Groups A-E)

New in DATA-4: 6 checks (Group G: GUT verification)

ENTRY COUNT:

Sections A-K (inherited from DATA-3): 123 entries

Section L (Cabibbo Doublet, STAGED): 6 entries

Section N (GUT parameters, Level 1): 17 entries

Total: 146 entries

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| ALL TESTS PASS

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DATA-4 DECLARATION:	
DATA-3 (123 entries, 32/32 checks) inherited.	
Finding 15 (lattice independence) formalized.	
6 Cabibbo Doublet parameters STAGED (entries 124-129).	
17 GUT/unification parameters added (Section N).	
6 new GUT verification checks added (Group G).	
Computation traceability map added (Group T).	
DATA-3 is retired.	
All future HOWL computation references DATA-4.	

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DATA-4 VERIFICATION COMPLETE

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 [HOWL-DATA-4-2026] Verified Integer Rational Database . Github: <https://github.com/ghowland/papers/tree/main/papers/DATA-4-2026>

[HOWL-DATA-1-2026] Cross-Domain Precision Data Inventory. Github: <https://github.com/ghowland/papers/tree/main/papers/DATA-1-2026>

[HOWL-DATA-2-2026] Integer Rational Representations in Q335. Github: <https://github.com/ghowland/papers/tree/main/papers/DATA/HOWL-DATA-2-2026>

[HOWL-DATA-3-2026] Verified Integer Rational Database. Github: <https://github.com/ghowland/papers/tree/main/papers/DATA-3-2026>